AQ4.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Match the vector spaces (with the usual scalar multiplication and vector addition as in $M_{3\times 3}(\mathbb{R})$) in column A with their bases in column B in Table : M2W4AQ2.

Choose the correct option.

- [ ] a $\rightarrow$ ii
- [ ] a $\rightarrow$ i
- [ ] b $\rightarrow$ iii
- [ ] c $\rightarrow$ i
- [ ] b $\rightarrow$ ii
- [ ] c $\rightarrow$ iii

No, the answer is incorrect.
Score: 0
Accepted Answers:
a $\rightarrow$ ii
b $\rightarrow$ iii
c $\rightarrow$ i

If S be a subset of $\mathbb{R}^5$ such that $span(S) = \mathbb{R}^5$, then what is the minimum number of elements possible in S?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 5
1 point
1 point
If S_1 is a maximal linear independent set and S_2 is a minimal spanning set of a vector space V, then which of the following option(s) is (are) true?

- [ ] For any $v \in S_1$, $S_1 \setminus \{v\}$ is a linearly independent.
- [ ] For any $v \in S_2$, $S_2 \setminus \{v\}$ is a spanning set of V.
- [ ] For any $v \in V \setminus S_1$, $S_1 \cup \{v\}$ is a linearly dependent.
- [ ] For any $v \in V$, $S_2 \cup \{v\}$ is a spanning set of V.

No, the answer is incorrect.
Score: 0
Accepted Answers:

For any $v \in S_1$, $S_1 \setminus \{v\}$ is a linearly independent.

For any $v \in V \setminus S_1$, $S_1 \cup \{v\}$ is a linearly dependent.

For any $v \in V$, $S_2 \cup \{v\}$ is a spanning set of V.

1 point

Consider the following subset of R^2 with usual addition and scalar multiplication as in R^2.

- $V_1 = \{(x, 0) \mid x \in R\}$
- $V_2 = \{(0, y) \mid y \in R\}$

Which of the following options are correct?

- [ ] Both V_1 and V_2 are subspaces of R^2.
- [ ] $V_1 \cap V_2$ is a subspace of R^2.
- [ ] $V_1 \cup V_2$ is a subspace of R^2.
- [ ] V_1 is a subspace of R^2, but V_2 is not.
- [ ] V_2 is a subspace of R^2, but V_1 is not.
- [ ] $\{(1, 0)\}$ is a basis of V_1.
- [ ] $\{(0, 1)\}$ is a basis of V_2.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Both V_1 and V_2 are subspaces of R^2.

$V_1 \cap V_2$ is a subspace of R^2.

$\{(1, 0)\}$ is a basis of V_1.

$\{(0, 1)\}$ is a basis of V_2.

1 point

Consider the following subset of R^2 with usual addition and scalar multiplication as in R^2.

- $V_1 = \{(x, 0) \mid x \in R\}$
- $V_2 = \{(2x, 0) \mid x \in R\}$

Which of the following options are correct?

- [ ] Both V_1 and V_2 are subspaces of R^2.
- [ ] $V_1 \cap V_2$ is a subspace of R^2.
- [ ] $V_1 \cup V_2$ is a subspace of R^2.
- [ ] V_1 is a subspace of R^2, but V_2 is not.
- [ ] V_2 is a subspace of R^2, but V_1 is not.
- [ ] $\{(1, 0)\}$ is a basis of V_1.
- [ ] $\{(1, 0)\}$ is a basis of V_2.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Both V_1 and V_2 are subspaces of R^2.
$V_1 \cap V_2$ is a subspace of R^2.
$V_1 \cup V_2$ is a subspace of R^2.
$\{(1,0)\}$ is a basis of V_1.
$\{(1,0)\}$ is a basis of V_2.

1 point
Let V be a vector space which is defined as follows:

$$V = \{(x, y, z, w) \mid x + z = y + w\} \subseteq R^4$$

with usual addition and scalar multiplication. Which of the following set forms a basis of V ?

- [ ] $\{(1,0,0,0), (0,1,0,0), (0,0,0,1)\}$.
- [ ] $\{(1,1,0,0), (0,1,-1,0), (0,-1,0,1)\}$.
- [ ] $\{(1,0,-1,0), (1,1,0,0), (1,0,0,1)\}$.
- [ ] $\{(1,1,0,0), (0,1,1,0), (0,-1,0,1)\}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(1,0,-1,0), (1,1,0,0), (1,0,0,1)\}$.
$\{(1,1,0,0), (0,1,1,0), (0,-1,0,1)\}$.

Level 2

1 point
Which of the following sets form a basis of the vector space of 2×2 lower triangular real matrices with usual matrix addition and scalar multiplication? (More than one option may be correct)

- [ ] $\left\{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\right\}$
- [ ] $\left\{\begin{bmatrix}1&0\\1&0\end{bmatrix}, \begin{bmatrix}0&0\\1&1\end{bmatrix}, \begin{bmatrix}1&0\\0&1\end{bmatrix}\right\}$
- [ ] $\left\{\begin{bmatrix}0&0\\1&0\end{bmatrix}\right\}$
- [ ] $\left\{\begin{bmatrix}-1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&-1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\right\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\left\{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\right\}$

$\left\{\begin{bmatrix}1&0\\1&0\end{bmatrix}, \begin{bmatrix}0&0\\1&1\end{bmatrix}, \begin{bmatrix}1&0\\0&1\end{bmatrix}\right\}$

$\left\{\begin{bmatrix}-1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&-1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\right\}$

If S is a linearly independent set in vector space $V = \{(x, y) \mid y = 2x, \text{ where } x, y \in R\}$ with usual addition and scalar multiplication as in R^2, then find the maximum possible cardinality of S.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1

1 point

1 point

If $\{v_1, v_2, v_3\}$ forms a basis of R^3, then which of the following are true?

- [ ] $\{v_1, v_2, v_1 + v_3\}$ forms a basis of R^3
- [ ] $\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of R^3.
- [ ] $\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of R^3.
- [ ] $\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of R^3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{v_1, v_2, v_1 + v_3\}$ forms a basis of R^3
$\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of R^3.
$\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of R^3.
$\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of R^3.

1 point

Which of the following options is(are) true?

- [ ] Any minimal spanning set of a vector space V must be a basis of V.
- [ ] Any maximal spanning set of a vector space V must be a basis of V.
- [ ] Any minimal linear independent set of vector space V must be a basis of V.
- [ ] Any maximal linear independent set of vector space V must be a basis of V.
- [ ] The basis of a vector space is unique.
- [ ] The number of elements in a basis of R^3 is 3.
- [ ] The number of elements in a basis of $M_{3\times 3}(R)$ is 3.
- [ ] There are infinite number of bases of R^3.
- [ ] Any subset of a minimal spanning set of V cannot be a spanning set.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Any minimal spanning set of a vector space V must be a basis of V.
Any maximal linear independent set of vector space V must be a basis of V.
The number of elements in a basis of R^3 is 3.

There are infinite number of bases of R^3 R3.
Any subset of a minimal spanning set of V V cannot be a spanning set.

Check Answers
Your score is: 0/10